

**LAB ASSIGNMENT**

Q-1 Decision Tree for Heart Disease Prediction

**Part A – Basic Implementation**

- Train a Decision Tree on the Heart Disease dataset.
- Report training and testing accuracy.
- Visualize the decision tree and interpret the first two splits.

**Part B – Model Evaluation**

- Print the **confusion matrix** and **classification report** (precision, recall, F1-score).
- Plot the **ROC curve** and report the **AUC score**.

**Part C – Hyperparameter Tuning**

- Vary max\_depth, min\_samples\_split, min\_samples\_leaf.
- Compare training vs testing accuracy → shows **overfitting vs underfitting**.
- Use **GridSearchCV** for automated hyperparameter tuning.

**Part D – Error Analysis**

- Find the patients (rows) that were misclassified by the decision tree.
- Compare their features with correctly classified patients.
- What patterns do you notice? (e.g., “younger patients with mild symptoms were misclassified”).

**Part E – Class Imbalance Check**

- Check class distribution (how many patients with disease vs without).
- If imbalanced → use metrics like **balanced accuracy** or apply resampling (SMOTE, undersampling).

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